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Challenging Status Quo with World's First Fully Automated Well Test Package, Allowing Ultra-High H₂S Reservoirs Evaluation

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Abstract

This paper details the successful deployment of a state-of-the-art, fully automated surface well testing platform in an ultra-high H₂S environment. The objective is to demonstrate how this medium-level automation solution enhances safety, improves efficiency, and minimizes environmental impact by automating critical tasks, reducing human exposure to hazardous conditions, and improving data accuracy and reliability.

Ultra-high H₂S environments pose significant challenges to traditional well testing operations due to the extreme toxicity and corrosiveness of hydrogen sulphide. Addressing these issues, efforts towards fully automated well testing began in 2020 with the deployment of the world's first electrical automated well testing choke manifold. This initiative evolved into the comprehensive system presented, enabling safe and efficient operations previously considered impractical.

Leveraging advanced automation technologies, remote operation capabilities, and enhanced safety layers, this paper presents the deployment of the world's first fully automated well test package, integrating all components from the choke manifold to the burners. This integrated system allows for the full control of produced oil, gas, and water automatically or from a safe remote location. The paper discusses specific challenges encountered in this ultra-high H₂S environment and how they were overcome through innovative design and deployment strategies. A detailed analysis of operational benefits demonstrates significantly optimized well test duration, reduced downtime, minimized human error, and optimized resource allocation.

In conclusion, this deployment in ultra-high H₂S environments represents a significant advancement in the scenarios and envelopes where well testing can be safely and efficiently performed. This paper presents the benefits from the first application of a fully automated well testing system encompassing the choke manifold, three-phase separator, transfer pump, surge tank, diverter manifold, and well testing burner deployed as a single, integrated package.

Introduction

The evaluation of hydrocarbon reservoirs is a critical process in the upstream oil and gas industry, providing essential data for reservoir characterization, production forecasting, and field development planning.

Well testing offers dynamic information about reservoir fluid properties, well productivity, and reservoir boundaries. However, these operations become significantly more complex and hazardous when dealing with reservoirs containing ultra-high concentrations of hydrogen sulphide (H₂S).

Hydrogen sulphide is a highly toxic, corrosive, and flammable gas that poses severe risks to personnel, equipment, and the environment. Traditional manual well testing operations in such environments necessitate extensive safety protocols, specialized equipment, and highly trained personnel, often leading to increased operational costs, extended testing durations, and inherent safety vulnerabilities. [Alkhoujah \(2024\)](#) mentions that human labour at high H₂S facilities involves the continuous use of self-contained breathing apparatus with operators needing to take breaks every hour. The manual manipulation of equipment, exposure to potential leaks, and reliance on human judgment under stressful conditions escalate the risk profile of such environments.

This paper addresses these challenges by introducing and detailing the successful deployment of a fully automated surface well testing package. This advanced system represents a significant shift from conventional manual operations, allowing for the safe and efficient evaluation of ultra-high H₂S reservoirs, which have historically been extremely dangerous to test. The initiative began in 2020 with the deployment of an electrical automated well testing choke manifold ([Alkhuwayr, 2022](#)). Since this date this new set of well testing equipment allowed the evaluation of those extreme condition reservoirs with improved HSE risk management, and here we will expand the scope for automation packages that have evolved into a comprehensive, integrated automation platform that includes all main equipment.

The primary objective of this paper is to present a case study demonstrating the capabilities and benefits of this automated well test package. We will highlight its key features, the technological advancements enabling its functionality, and its profound impact on safety, operational efficiency, data accuracy, and environmental stewardship in extreme H₂S environments.

The subsequent sections of this paper will cover the inherent challenges of well testing in ultra-high H₂S environments, the methodological approach to designing and deploying the automated system, the specific components integrated into the fully automated package, the operational and safety enhancements observed, and finally, the conclusions and suggestions for future applications of this innovative technology. By addressing these critical aspects, this paper aims to provide a comprehensive overview of the advancements in well testing automation and their implications for the oil and gas industry.

Challenges of Well Testing in Ultra-High H₂S Environments

OSHA defines 100 ppm of H₂S as immediately dangerous to life and health, while different sources classify wells by reservoir concentrations. [Jauregui \(2023\)](#) defines ultra-high H₂S reservoirs as those containing >10% and up to 40% H₂S, the focus of this work. Such environments present significant challenges across all aspects of well testing operations:

Severe Safety Risks to Personnel

- **Extreme Toxicity:** H₂S is lethal at low concentrations, with exposure potentially causing rapid unconsciousness, respiratory failure, and death. This necessitates stringent use of personal protective equipment (PPE), continuous gas monitoring, comprehensive emergency response plans, and extensive training.
- **Corrosion:** H₂S promotes sulphide stress cracking (SSC) and hydrogen embrittlement, particularly in wet conditions, requiring NACE-compliant metallurgy, frequent inspections, and specialized equipment ([NACE, 2021](#)).
- **Flammability:** H₂S forms explosive mixtures with air, demanding strict ignition source control and robust fire prevention measures.

Operational Inefficiencies and Limitations

- **Extended Test Duration:** Manual operations require reduced flow periods, frequent safety stand-downs, and limited personnel exposure, prolonging test timelines.
- **Increased Manpower and Logistics:** Additional personnel are needed for safety watch, emergency readiness, and redundant operations, while H₂S-trained teams and specialized equipment increase logistical complexity (Alkhoujah, 2024).
- **Data Quality Concerns:** Hazardous conditions, fatigue, and manual intervention can compromise data accuracy and reliability.
- **Equipment Availability:** Conventional well test equipment may not be rated for ultra-high H₂S service or may require extensive modification, increasing lead times and costs.
- **Environmental Impact:** Managing H₂S flaring to limit SO₂ emissions requires specialized burners and continuous monitoring.

Regulatory and Environmental Compliance

- **Regulatory Requirements:** H₂S handling and flaring are subject to strict environmental and safety regulations, requiring robust monitoring, mitigation measures, and permitting (OSHA, 2024).
- **Social License:** Operations near communities depend on demonstrated safety performance and environmental responsibility.

These challenges highlight the need for innovative solutions that reduce risk, improve operational efficiency, and ensure regulatory compliance. Traditional well testing methods, while effective in conventional reservoirs, are increasingly unsuited to ultra-high H₂S applications.

Methodology and System Design

To address the challenges of ultra-high H₂S well testing, a phased approach was adopted to develop and deploy a bespoke fully automated surface well testing package. This method allowed gradual client engagement, operator training, and incorporation of lessons learned before final system release.

The methodology focused on integrating advanced process control, industry-leading instrumentation, and automation rated for hazardous environments, evolving from the prior manually operated well testing systems.

Design Principles

Key design principles for the fully automated package included:

- **Targeted Medium Automation Level:** Designed to significantly reduce human intervention at the well site—especially in H₂S exposure zones—while preserving critical human decision-making where needed (see Figure 1, Franklin 2024). A higher automation level well testing plant can analyze information that enables the execution of tasks with minimal (or no) human intervention, including detection of anomalies in the dataset that deviates from the "design of service" or the "equipment operating envelope,".
- **Enhanced Safety:** Incorporating layered safety interlocks and continuous flowback monitoring to eliminate hazardous manual tasks. Risk is reduced to the lowest achievable without excessive cost, effort, or time, and while not reducing the functionality or operability of equipment.
- **Robustness and Reliability:** Employing NACE MR0175/ISO 15156-compliant materials (NACE, 2021; API, 2018), electrical actuation, monitoring, and redundancy to reduce process control failures.

- **Data Integrity:** Utilizing high-fidelity sensors and automated data acquisition for accurate, real-time data.
- **Remote Operability:** Enabling full control and monitoring from safe remote locations.
- **Modularity:** Designing scalable, efficiently deployable system components.

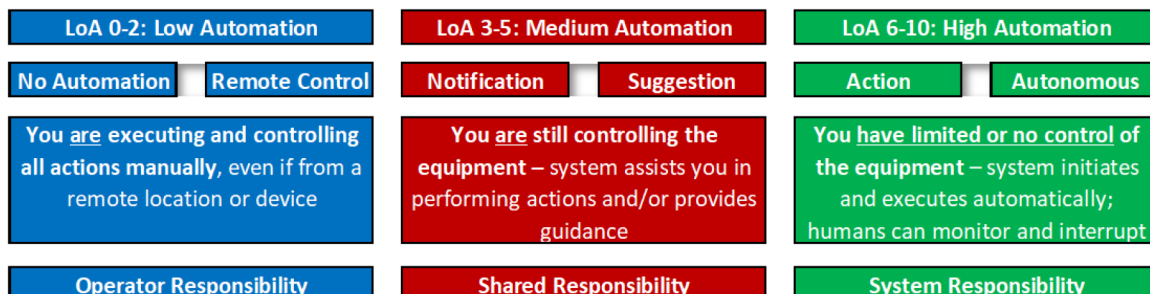


Figure 1—Levels of automation for surface well testing systems (Source: Halliburton)

Phased Automation Development

The automation pathway began in 2020 with the deployment of the world's first electrically actuated automated choke manifold (Alkhuwayr, 2022), validating automated flow control in ultra-high H₂S environments. Insights from this phase guided the development of the fully integrated well testing package.

Components of the Fully Automated Well Test Package

The integrated system includes all major well test equipment, each outfitted with sensors, actuators, and communication modules for automated operation:

- **Choke Manifold:** The pioneering component providing precise remote control of flow rates and wellhead pressure with electric actuators and continuous condition monitoring (Figure 2).
- **Three-Phase Separator:** Designed for automated level control of oil, water, and gas with real-time flow, temperature, and pressure measurements. It autonomously stabilizes under varying well conditions (e.g., slug flow, choke changes), adding safety layers (Figure 3).
- **Transfer Pump:** Automated start/stop and speed control for fluid transfer, reducing manual handling with added safety.
- **Surge Tank:** Automated level and pressure management to buffer transient flows and enable automated metering factor and shrinkage calculations.
- **Diverter Manifold:** Automated valves directing flow between separator, surge tank, burner, and disposal pits, facilitating safe realignment during changing conditions.
- **Well Testing Burner & Flare:** Automated ignition, flame monitoring, and shutdown systems ensure efficient combustion of oil, gas, and H₂S with sensors confirming complete conversion to SO₂.

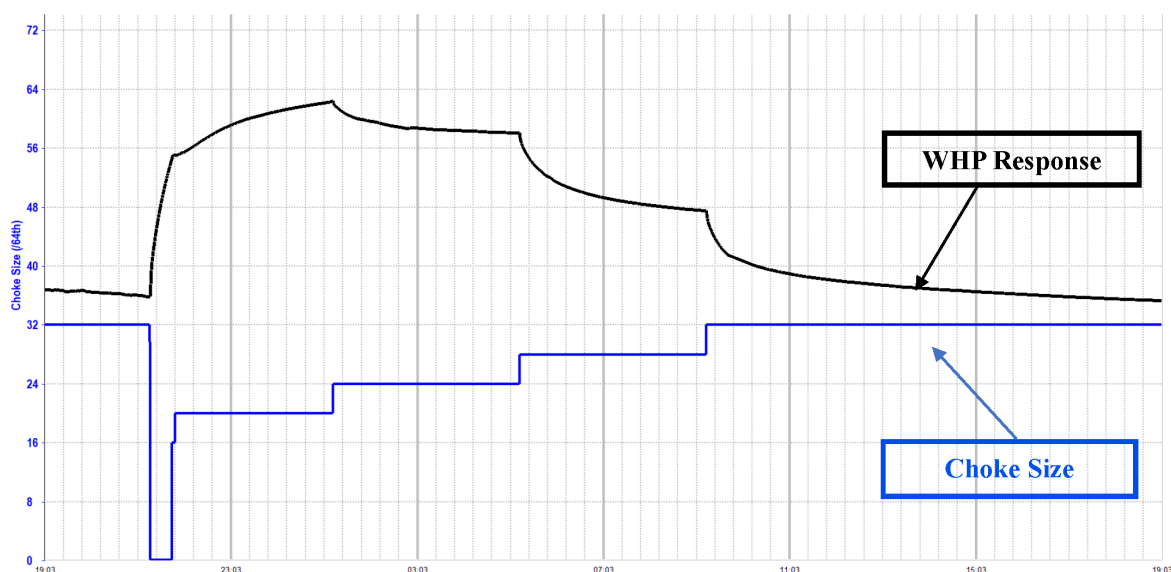


Figure 2—Precise Choke changes without pressure spike

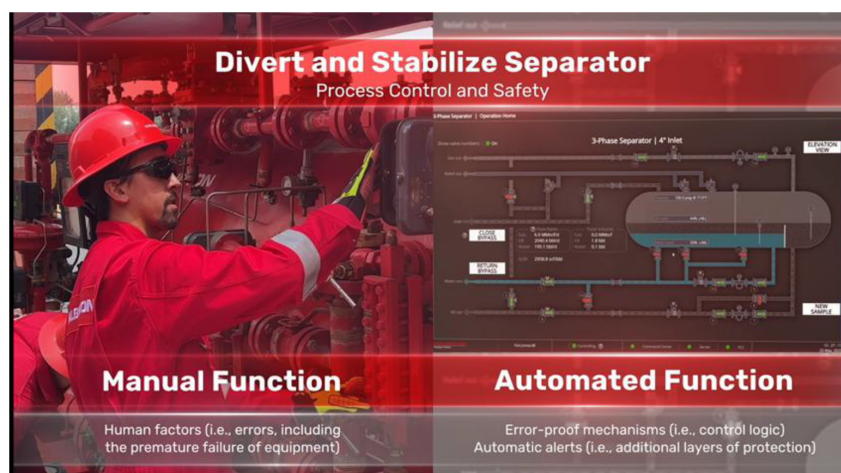


Figure 3—Comparison of operating a manual three-phase separator and an automated (Source: Halliburton).

Process Control and Safety Systems

- The package is controlled by a robust PLC-based process control system that processes sensor data, executes automated workflows, and displays alarms for operator oversight.
- Secure communication protocols ensure reliable data exchange between field instruments and the remote command center, supported by four interdependent safety layers.
- A Human-Machine Interface (HMI) provides real-time visualization, control, and alarm management from safe zones via computers or tablets.
- An independent Safety Instrumented System (SIS) employs Electrical Emergency Shutdown (E-ESD) capabilities to rapidly isolate the well in emergencies, operable both locally and remotely. Safety instruments include high/low-pressure sensors, temperature monitors, and gas detectors.
- Process and safety systems are independent to reduce common-cause failures and increase overall protection.
- Continuous H_2S gas detectors are strategically deployed and integrated with control and safety systems for automated alarms and shutdowns. Fire and gas detection systems complement these protections.

- Gas and liquid analysers, automated purging, and remote blowdown capabilities minimize H₂S residues and enable rapid depressurization if required, key for continuous monitoring of equipment under extreme conditions.
- Materials strictly comply with NACE MR0175/ISO 15156 standards, employing alloys such as Inconel 725, Hastelloy, stainless steel, and carbon steel, with design emphasis on eliminating elastomeric seals in gas-contacting areas.

This systematic approach resulted in an integrated, intelligent well testing package capable of safe, reliable, and efficient operation in ultra-high H₂S environments.

Case Study: Deployment in an Ultra-High H₂S Environment

This section presents the application and performance of the fully automated well test package in a challenging ultra-high H₂S reservoir in 2024. The system enabled safe, efficient testing of a major gas field with H₂S concentrations up to 40%, achieving **zero HSE incidents** and **zero non-productive time (NPT)**.

Specific Challenge

Traditional well testing for hazardous wells often involves complex rig-ups, extended flow periods to stabilize conditions, and complicated logistics for managing and disposing of flowback fluids. These factors prolong testing duration, increase rig and personnel costs, and delay production. The operator sought a solution that would eliminate added cleanup and testing time **without compromising data quality or safety**.

Implementation and Strategy

A fully integrated electrical automated package was deployed, designed to enhance process safety, minimize operational risk, and reduce environmental impact—even under extreme H₂S conditions.

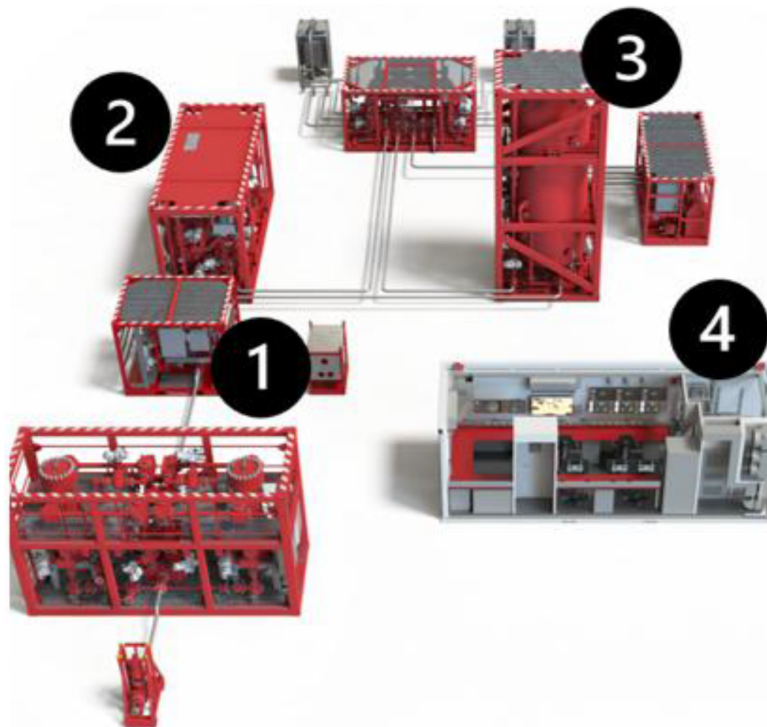


Figure 4—Overview of the automated package (Source: Halliburton)

Key components

1. **Electrical Choke Manifold:** Provided precise, automated well control without personnel exposure or disruptive events.
2. **Electrical Three-Phase Separator:** Enabled unmanned, separation with automated response to transient well conditions, maintaining process stability and data accuracy. Provides enhanced process control with robustness in measurement accuracy, reducing the whole well testing risk.
3. **Electrical Disposal Package:** Included surge tank, transfer pump, diverter manifold, and burner—fully automated to reduce rig time and personnel presence in red zones as shown on results chapter.
4. **Command Center:** Centralized monitoring and control from a safe location, with real-time visibility via HMI screens, tablets, or remote operation centers.

The modular, plug-and-play design—with no-hammer pipework and a five-way automated diverter manifold—streamlined rig-up, minimized footprint, and enabled **real-time data acquisition** for immediate well performance insights.

Results

Transformative Impact on Safety. One of the key design requirements, a drastic reduction in personnel exposure to lethal H₂S gas was observed. Automation reduced manual interventions at the wellsite, and the integrated design minimized break of containment and potential leak paths, enhancing overall operational safety.

Many manual red-zone tasks were eliminated. The system shifted safety philosophy from **managing exposure** to **eliminating exposure** where feasible. Added layers—such as E-ESD, continuous gas detection, and integrated shutdowns—further safeguarded operations.

Operational Performance Summary.

- **17% reduction in well test duration** — achieved through optimized choke operations, automated separator stabilization, and rapid fluid diversion. No productive time was recorded as the system provides process alarms to keep equipment under operational envelope all the time.
- **29% reduction in red-zone personnel** — with crew reallocated from hazardous tasks to higher-value data verification activities.
- **238 tons of CO₂ emissions avoided** — due to shorter test duration and more efficient burning.
- **Optimized well total cost of ownership** — from savings in rig time, crew costs, and service requirements.
- **High-quality, real-time data** — critical for rapid reservoir evaluation and decision-making.

Operational Efficiency Gains.

- **More data, faster:** Enhanced acquisition allows richer datasets within the same timeframe.
- **Reduced rig/unit time:** Faster tests minimized rig idle time waiting on results.
- **Accelerated decision-making:** Real-time, high-fidelity data enabled quicker, better-informed development choices for reservoir engineers and asset managers.

Superior Data Quality and Reliability. Automated systems delivered accurate, consistent data without manual transcription errors. Continuous data streams enhanced pressure transient analysis, fluid characterization, and reservoir model calibration for more robust management decisions.

Economic Impact. Though requiring higher upfront investment, automation provided long-term value through:

- Reduced operational expenditure (OPEX) via leaner staffing and lower exposure-related costs.
- Minimized NPT for rigs and well test units.
- Avoidance of HSE incidents and associated penalties.
- Unlocking production from ultra-high H₂S fields that might otherwise remain uneconomic.

Future Enhancements. Potential future developments include:

- Tighter integration with upstream and downstream data systems for end-to-end visibility.
- Real-time optimization via predictive analytics and machine learning.
- Progress toward fully autonomous operation for routine tasks.
- Advanced sensors for even harsher environments.

Conclusions

This paper has presented a pioneering case study on the successful **design, development, and deployment of the world's first fully automated well test package** for ultra-high H₂S environments. The system marks a major advancement in surface well testing technology, directly addressing the critical challenges of safety, efficiency, and data integrity when handling hazardous reservoir fluids.

The integrated automated package—including the choke manifold, three-phase separator, transfer pump, surge tank, diverter manifold, and well testing burner—delivers complete automated control and monitoring of well test operations. This has fundamentally shifted the operational approach from manual intervention to automated execution, **dramatically reducing human exposure to lethal H₂S concentrations** and mitigating associated risks.

The operational benefits demonstrated through this deployment are substantial and include:

- **Optimized well test duration and reduced non-productive time**
- **Enhanced data accuracy, consistency, and integrity**
- **Minimized human error and manual intervention in hazardous zones**
- **Significant economic and environmental gains** through reduced rig time, lower OPEX, and decreased emissions

This successful deployment proves the **feasibility, reliability, and value of advanced automation** in even the most extreme testing conditions. It expands the operational envelope for safe and efficient well testing in challenging environments and establishes a new benchmark for safety, operational excellence, and technological innovation in the industry.

Future Work

Building on the success of this initial deployment, several pathways are identified to further enhance the capabilities and value of fully automated well testing in hazardous environments:

- **Expansion of Remote Operations:** The system's architecture enables seamless integration with Remote Operations Centers (ROCs). This will allow further reduction of on-site personnel, enhance

collaboration with technical teams in town, and provide higher levels of oversight and operational efficiency.

- **Higher-Level Automation with Advanced Analytics and Predictive Maintenance:** Future enhancements will focus on integrating advanced analytics platforms and machine learning models. These will support predictive maintenance strategies, early detection of equipment anomalies, optimization of test parameters in real time, and potentially autonomous adjustment of operational settings to further reduce risk and enhance data quality.
- **Integration with Digital Twin Technology:** Developing a comprehensive digital twin of the well test package will enable real-time simulation, predictive modelling, scenario analysis, and virtual operator training. This will enhance operational predictability, decision-making, and safety assurance.
- **Establishing Automation as Standard for Complex Operations:** The demonstrated benefits of automation in ultra-high H₂S environments provide a strong case for adopting automated well test packages as the standard safeguard in future high-risk operations. This approach not only eliminates unnecessary personnel exposure to extreme HSE hazards but also opens new possibilities for evaluating and producing from reservoirs previously deemed too risky or uneconomical to develop.

Acknowledgment

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Nomenclature

H ₂ S	: Hydrogen Sulfide
SO ₂	: Sulfur Dioxide
PPE	: Personal Protective Equipment
SSC	: Sulfide Stress Cracking
NACE	: National Association of Corrosion Engineers
API	: American Petroleum Institute
OSHA	: Occupational Safety and Health Administration
PLC	: Programmable Logic Controller
HMI	: Human-Machine Interface
E-ESD	: Electrical Emergency Shut Down
OPEX	: Operational Expenditure
NPT	: Non-Productive Time
HSE	: Health Safety and Environment
ROC	: Remote Operation Center

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